

Quiz 2

Instructions: There are totally 2 problems. You must show all of your work to receive a credit for a correct answer.

Problem 1. (5 points) Find a general solution (explicit if convenient) for the first-order differential equation $\frac{dy}{dx} = 2x \sec(y)$.

$$\frac{dy}{dx} = 2x \sec(y)$$

$$\frac{dy}{dx} = \frac{2x}{\cos y}$$

$$\cos y \, dy = 2x \, dx$$

$$\int \cos y \, dy = \int 2x \, dx$$

$$\sin y = x^2 + C \Rightarrow y = \arcsin(x^2 + C)$$

General solution

Problem 2. (5 points) Find the explicit particular solution for the initial value problem

$$\frac{dy}{dx} = y - 2xy, \quad y(1) = -4.$$

$$\frac{dy}{dx} = y(1-2x)$$

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$$\frac{1}{y} dy = (1-2x) dx$$

$$\int \frac{1}{y} dy = \int (1-2x) dx$$

$$\ln|y| = x - x^2 + C$$

$$|y| = e^{x-x^2+C}$$

$$= Ae^{x-x^2} \quad (A > 0)$$

Thus

$$y = Ae^{x-x^2} \quad \text{or} \quad y = -Ae^{x-x^2}$$

(>0) (≤ 0)

Since $y(1) = -4 < 0$, we know

$$y = -Ae^{x-x^2}$$

and

$$-Ae^{1-1^2} = -4$$

$$\Rightarrow -A = -4 \Rightarrow A = 4$$

Thus the particular solution is

$$y = -4e^{x-x^2}$$